

- This intuition can be formalized, after some logical simplification, by the following explanation closure axiom:

$$\neg broken(x, s) \wedge broken(x, do(a, s)) \supset \\ (\exists r) a = drop(r, x) \wedge fragile(x) \vee \\ (\exists b) \{a = explode(b) \wedge nexto(b, x, s)\}.$$

- Similarly, (4) yields the following explanation closure axiom:

$$broken(x, s) \wedge \neg broken(x, do(a, s)) \supset \\ (\exists r) a = repair(r, x).$$

Aside: Normal Forms for Effect Axioms

- In the previous example, we rewrote one or more positive effect axioms as the single, logically equivalent positive effect axiom with the following syntactic normal form:

$$\gamma_F^+(\vec{x}, a, s) \supset F(\vec{x}, do(a, s)),$$

Similarly, we rewrote one or more negative effect axioms in the normal form:

$$\gamma_F^-(\vec{x}, a, s) \supset \neg F(\vec{x}, do(a, s)).$$

Here, $\gamma_F^+(\vec{x}, a, s)$ and $\gamma_F^-(\vec{x}, a, s)$ are first order formulas whose free variables are among $\vec{x}, a, s$.

The automatic generation of frame axioms appealed to these normal forms for the effect axioms.

- *Transformation of Effect Axioms to Normal Form:*
 1. Each of the given positive effect axioms has the form:

$$\phi_F^+ \supset F(\vec{t}, do(\alpha, s)).$$

Here α is an action term (e.g. $\text{pickup}(x)$, $\text{put}(A,y)$) and the $\vec{t}$ are terms.

Write this in the following, logically equivalent form:

$$a = \alpha \wedge \vec{x} = \vec{t} \wedge \phi_F^+ \supset F(\vec{x}, do(a, s)). \quad (5)$$

Here, $\vec{x} = \vec{t}$ abbreviates $x_1 = t_1 \wedge \cdots \wedge x_n = t_n$, and $\vec{x}$ are new variables, distinct from one another and distinct from any occurring in the original effect axiom.

2. Suppose $y_1, \dots, y_m$ are all the variables occurring in the original effect axiom. Then (5) is itself logically equivalent to:

$$(\exists y_1, \dots, y_m)[a = \alpha \wedge \vec{x} = \vec{t} \wedge \phi_F^+] \supset F(\vec{x}, do(a, s)).$$

3. So, each positive effect axiom for fluent F can be written in the logically equivalent form:

$$\Psi_F \supset F(\vec{x}, do(a, s)),$$

where Ψ_F is a formula whose free variables are among $\vec{x}, a, s$.

Do this for each of the k positive effect axiom for F , to get:

$$\begin{aligned} \Psi_F^{(1)} &\supset F(\vec{x}, do(a, s)), \\ &\vdots \\ \Psi_F^{(k)} &\supset F(\vec{x}, do(a, s)). \end{aligned}$$

4. Write these k sentences as the single, logically equivalent

$$[\Psi_F^{(1)} \vee \dots \vee \Psi_F^{(k)}] \supset F(\vec{x}, do(a, s)).$$

This is the normal form for the positive effect axioms for fluent F .

5. Similarly, compute the normal form for the negative effect axioms for fluent F .

- For those of you who know about semantics for logic programming: The above transformation to normal form is very similar (but not identical) to the preliminary transformation of logic program clauses, in preparation for computing a program's *completion*.

- *Example:*

Suppose the following are all the positive effect axioms for fluent *tired*:

$$\begin{aligned} & tired(Jack, do(walk(A, B), s)), \\ & \neg marathonRunner(y) \wedge distance(u, v) > 2km \supset \\ & \quad tired(y, do(run(u, v), s)). \end{aligned}$$

Their normal form is:

$$\begin{aligned} & \{[a = walk(A, B) \wedge x = Jack] \vee [(\exists u, v, y). a = run(u, v) \wedge \\ & \quad \neg marathonRunner(y) \wedge distance(u, v) > 2km \wedge x = y]\} \\ & \supset tired(x, do(a, s)). \end{aligned}$$

By properties of equality and existential quantification, this simplifies to:

$$\begin{aligned} & \{[a = walk(A, B) \wedge x = Jack] \vee [(\exists u, v). a = run(u, v) \wedge \\ & \quad \neg marathonRunner(x) \wedge distance(u, v) > 2km]\} \\ & \supset tired(x, do(a, s)). \end{aligned}$$

A Simple Solution: The General Case

- The previous example obviously generalizes. We suppose given, for each fluent F , the following two normal form effect axioms:
Positive Normal Form Effect Axiom for Fluent F

$$\gamma_F^+(\vec{x}, a, s) \supset F(\vec{x}, do(a, s)). \quad (6)$$

Negative Normal Form Effect Axiom for Fluent F

$$\gamma_F^-(\vec{x}, a, s) \supset \neg F(\vec{x}, do(a, s)). \quad (7)$$

Here, $\gamma_F^+(\vec{x}, a, s)$ and $\gamma_F^-(\vec{x}, a, s)$ are first order formulas whose free variables are among $\vec{x}, a, s$.

- *Causal Completeness Assumption:*
Axioms (6) and (7), respectively, characterize all the conditions under which action a can lead to F becoming true (respectively, false) in the successor situation. They are all the causal laws for the fluent F .
- Hence, if F 's truth value changes from *false* to *true* as a result of doing a , then $\gamma_F^+(\vec{x}, a, s)$ must be *true*; similarly, if F 's truth value changes from *true* to *false*.
- This informally stated assumption can be represented axiomatically by the following:

Explanation Closure Axioms

$$F(\vec{x}, s) \wedge \neg F(\vec{x}, do(a, s)) \supset \gamma_F^-(\vec{x}, a, s), \quad (8)$$

$$\neg F(\vec{x}, s) \wedge F(\vec{x}, do(a, s)) \supset \gamma_F^+(\vec{x}, a, s). \quad (9)$$

- To make this work, we need *Unique Names Axioms for Actions*.

For distinct action names A and B ,

$$A(\vec{x}) \neq B(\vec{y}).$$

Identical actions have identical arguments:

$$A(x_1, \dots, x_n) = A(y_1, \dots, y_n) \supset x_1 = y_1 \wedge \dots \wedge x_n = y_n.$$

- **Result 1** *Let T be a first-order theory that entails*

$$\neg(\exists \vec{x}, a, s). \gamma_F^+(\vec{x}, a, s) \wedge \gamma_F^-(\vec{x}, a, s).$$

Then T entails that the general effect axioms (6) and (7), together with the explanation closure axioms (8) and (9), are logically equivalent to:

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee F(\vec{x}, s) \wedge \neg \gamma_F^-(\vec{x}, a, s). \quad (10)$$

- The requirement that

$$\neg(\exists \vec{x}, a, s). \gamma_F^+(\vec{x}, a, s) \wedge \gamma_F^-(\vec{x}, a, s)$$

be entailed by the background theory T simply guarantees the integrity of the effect axioms (6) and (7); under these circumstances, it will be impossible for both $F(\vec{x}, do(a, s))$ and $\neg F(\vec{x}, do(a, s))$ to be simultaneously derived. Notice that by the unique names axioms for actions, this condition is satisfied by the example treating the fluent *broken* above.

- Call formula (10) the *successor state axiom for fluent F* .

- For the above example, the successor state axiom for *broken* is:

$$\begin{aligned} broken(x, do(a, s)) \equiv & \\ & (\exists r)\{a = drop(r, x) \wedge fragile(x)\} \vee \\ & (\exists b)\{a = explode(b) \wedge nexto(b, x, s)\} \vee \\ & broken(x, s) \wedge \neg(\exists r)a = repair(r, x). \end{aligned}$$

A Simple Solution: Summary

- Our proposed solution yields the following axioms:

1. Successor state axioms: for each fluent F ,

$$F(\vec{x}, do(a, s)) \equiv \gamma_F^+(\vec{x}, a, s) \vee F(\vec{x}, s) \wedge \neg \gamma_F^-(\vec{x}, a, s).$$

2. For each action A , a single action precondition axiom of the form:

$$Poss(A(\vec{x}), s) \equiv \Pi_A(\vec{x}, s).$$

3. Unique names axioms for actions.

- Ignoring the unique names axioms (whose effects can be compiled), this axiomatization requires $\mathcal{F} + \mathcal{A}$ axioms in total, compared with the $2 \times \mathcal{A} \times \mathcal{F}$ explicit frame axioms that would otherwise be required.
- But maybe we get fewer axioms at the expense of prohibitively long successor state axioms.
 - A successor state axiom's length is roughly proportional to the number of actions which affect the truth value of the fluent F .
 - The intuition leading to the frame problem is that most actions don't affect F . So few actions affect it. So its successor state axiom is short.
- The conciseness and perspicuity of this axiomatization relies on three things:
 1. Quantification over actions.

2. The assumption that relatively few actions affect a given fluent.
3. The Generalized Completeness Assumption.

Deductive Planning with the Situation Calculus

- Planning: find a sequence of actions which, if performed in a world with an axiomatized initial situation, will lead to a world situation in which some goal statement will be true.

- **Example:**

Action precondition axioms:

$$\begin{aligned} Poss(pickup(r, x), s) \equiv & robot(r) \wedge \\ & [(\forall z) \neg holding(r, z, s)] \wedge nexto(r, x, s), \end{aligned} \quad (11)$$

$$Poss(walk(r, y), s) \equiv robot(r). \quad (12)$$

$$Poss(drop(r, x), s) \equiv robot(r) \wedge holding(r, x, s). \quad (13)$$

Effect axioms:

$$\begin{aligned} & holding(r, x, do(pickup(r, x), s)), \\ & \neg holding(r, x, do(drop(r, x), s)), \\ & nexto(r, y, do(walk(r, y), s)), \\ & nexto(r, y, s) \supset nexto(x, y, do(drop(r, x), s)), \\ & y \neq x \supset \neg nexto(r, x, do(walk(r, y), s)), \\ & onfloor(x, do(drop(r, x), s)), \\ & \neg onfloor(x, do(pickup(r, x), s)). \end{aligned}$$

Solve the frame problem:

Successor state axioms:

$$\begin{aligned} holding(r, x, do(a, s)) \equiv & a = pickup(r, x) \vee \\ & holding(r, x, s) \wedge a \neq drop(r, x), \end{aligned} \quad (14)$$

$$\begin{aligned}
nexto(x, y, do(a, s)) &\equiv a = walk(x, y) \vee \\
&(\exists r)[nexto(r, y, s) \wedge a = drop(r, x)] \vee \\
&nexto(x, y, s) \wedge \neg(\exists z)[a = walk(x, z) \wedge z \neq y],
\end{aligned} \tag{15}$$

$$\begin{aligned}
onfloor(x, do(a, s)) &\equiv (\exists r)a = drop(r, x) \vee \\
&onfloor(x, s) \wedge \neg(\exists r)a = pickup(r, x).
\end{aligned} \tag{16}$$

Unique names axioms for actions:

$$\begin{aligned}
pickup(r, x) &\neq drop(r', y), \\
pickup(r, x) &\neq walk(r', y), \\
walk(r, y) = walk(r', y') &\supset r = r' \wedge y = y', \\
etc.
\end{aligned}$$

Initial situation:

$$\begin{aligned}
chair(C), \quad robot(R), \quad nexto(R, A, S_0), \\
(\forall z)\neg holding(R, z, S_0).
\end{aligned} \tag{17}$$

Deductive Planning (Continued)

- Some derived facts:

$$holding(R, A, do(pickup(R, A), S_0)). \quad (18)$$

From (17), (18), (14) and unique names for actions,

$$holding(R, A, do(walk(R, y), do(pickup(R, A), S_0))). \quad (19)$$

From (17), (19) and (15),

$$nexto(A, y, do(drop(R, A), do(walk(R, y), do(pickup(R, A), S_0)))). \quad (20)$$

From (17), (16) and (19),

$$onfloor(A, do(drop(R, A), do(walk(R, y), do(pickup(R, A), S_0)))). \quad (21)$$

- Suppose we want to derive:

$$(\exists s).nexto(A, B, s) \wedge onfloor(A, s).$$

i.e. there is a world situation in which A is next to B and A is on the floor.

- The above is a constructive proof of this sentence, with

$$s = do(drop(R, A), do(walk(R, B), do(pickup(R, A), S_0))).$$

- A *plan* to get A onto the floor next to B .

Deductive Planning (Continued)

- Key idea: Plan synthesis as a side effect of theorem proving (Green, 1969, early versions of Shakey the SRI robot).

$$Axioms \models (\exists s)G(s).$$

- Any binding for s as a side effect of a proof is a plan guaranteed to yield a world situation satisfying the goal G .
- c.f. Prolog.
- Slight problem: Can also prove

$$(\exists s).nexto(A, B, s) \wedge onfloor(A, s),$$

with

$$s = do(drop(C, A), do(walk(C, B), do(pickup(C, A), S_0))).$$

- The first plan, in which the robot R does the work, conforms to the action precondition axioms, while the second plan does not; according to these axioms, robots can pick things up, and go for walks, but chairs cannot.
- The robot's plan is *executable* according to the action precondition axioms, meaning that one can prove, from the axioms, that:

$$\begin{aligned} & Poss(pickup(R, A), S_0) \wedge \\ & Poss(walk(R, B), do(pickup(R, A), S_0)) \wedge \\ & Poss(drop(R, A), do(walk(R, B), do(pickup(R, A), S_0))). \end{aligned}$$

There is no proof that this sequence of actions, as performed by the chair, is executable.

- **Official characterization of planning in the sitcalc:**

Relative to some background axioms, establish that

$$Axioms \vdash (\exists s).executable(s) \wedge G(s).$$

AN APPLICATION: FORMALIZING DATABASE UPDATE TRANSACTIONS